API-Design – LookUpStock

Stand: 30.09.2026 · Basis-URL /api/v1 · Datenformat JSON · Ab Sprint 2 Anmeldung per Header Authorization: Bearer <token>

Fehler haben immer das Format { "error": { "code", "message", "details" } }. message ist ein deutscher Text zur direkten Anzeige im Frontend. Gewichtungen in Prozent (Summe 100), Anteile in Ergebnissen als Dezimalzahl (0.31 = 31 %).

Auth (US-01)

Method	Ressource	Data In	Data Out

POST	`/auth/register`
	Konto anlegen · Sprint 2
		`{`
		`  "email": "anna@example.com",`
		`  "password": "geheim123"`
		`}`
			201 Created
			`{`
			`  "id": 3,`
			`  "email": "anna@example.com"`
			`}`
			400 Bad Request
			`{`
			`  "error": {`
			`  "code": "VALIDATION_ERROR",`
			`  "message": "Das Passwort muss mindestens 8 Zeichen haben."`
			`  }`
			`}`
			409 Conflict
			`{`
			`  "error": {`
			`  "code": "CONFLICT",`
			`  "message": "Diese E-Mail ist bereits registriert."`
			`  }`
			`}`

POST	`/auth/login`
	Anmelden · Sprint 2
		`{`
		`  "email": "anna@example.com",`
		`  "password": "geheim123"`
		`}`
			200 OK
			`{`
			`  "token": "eyJhbGciOiJIUzI1NiJ9...",`
			`  "expiresAt": "2026-10-13T10:00:00Z"`
			`}`
			401 Unauthorized
			`{`
			`  "error": {`
			`  "code": "UNAUTHORIZED",`
			`  "message": "E-Mail oder Passwort ist falsch."`
			`  }`
			`}`

GET	`/auth/me`	Header: Authorization: Bearer <token>	**200 OK**

Method	Ressource	Data In	Data Out

	Angemeldeten Nutzer abrufen · Sprint 2		{
			"id": 3,
			"email": "anna@example.com"
			}
			401 Unauthorized
			{
			"error": {
			"code": "UNAUTHORIZED",
			"message": "Bitte melde dich an."
			}
			}

Anlagen (US-02)

Method	Ressource	Data In	Data Out

GET	`/assets/search?q=msci&limit=10`
	Anlagen suchen · Sprint 1
		Query: q (Pflicht, mind. 2 Zeichen: Ticker, ISIN oder Name), limit (optional, Standard 10, max. 25). Kein Body.	**200 OK**
			{
			"query": "msci",
			"results": [
			{
			"id": 17,
			"symbol": "EUNL",
			"isin": "IE00B4L5Y983",
			"name": "iShares Core MSCI World UCITS ETF",
			"exchange": "XETRA",
			"currency": "EUR",
			"type": "ETF"
			}
			]
			}
			200 OK (keine Treffer)
			{
			"query": "xyzabc",
			"results": []
			}
			400 Bad Request
			{
			"error": {
			"code": "VALIDATION_ERROR",
			"message": "Bitte mindestens 2 Zeichen eingeben."
			}
			}

Method	Ressource	Data In	Data Out

			503 Service Unavailable
			{
			"error": {
			"code": "UPSTREAM_UNAVAILABLE",
			"message": "Kursdaten sind gerade nicht verfügbar."
			}
			}

GET	/assets/{assetId}
	Details einer Anlage · Sprint 1
		Pfad: assetId, z. B. /assets/17. Kein Body.	**200 OK**
			{
			"id": 17,
			"symbol": "EUNL",
			"isin": "IE00B4L5Y983",
			"name": "iShares Core MSCI World UCITS ETF",
			"exchange": "XETRA",
			"currency": "EUR",
			"type": "ETF"
			}
			404 Not Found
			{
			"error": {
			"code": "NOT_FOUND",
			"message": "Anlage nicht gefunden."
			}
			}

Portfolios (US-03, US-06)

Method	Ressource	Data In	Data Out

POST	/portfolios
	Portfolio anlegen · Sprint 1
		{
		"name": "Mein Weltportfolio",
		"positions": [
		{
		"assetId": 17,
		"weight": 70
		},
		{
		"assetId": 23,
		"weight": 30
		}
		]
		}
			201 Created
			{
			"id": 5,
			"name": "Mein Weltportfolio",
			"createdAt": "2026-09-30T10:12:00Z",
			"updatedAt": "2026-09-30T10:12:00Z",
			"positions": [
			{
			"assetId": 17,
			"symbol": "EUNL",
			"name": "iShares Core MSCI World",
			"weight": 70
			},

Method	Ressource	Data In	Data Out

		Ab Sprint 2 zusätzlich Header: Authorization: Bearer <token>	<pre>{
			"assetId": 23,
			"symbol": "IS3N",
			"name": "iShares Core MSCI EM IMI",
			"weight": 30
			}
			]
			}</pre>**400 Bad Request**<pre>{
			"error": {
			"code": "VALIDATION_ERROR",
			"message": "Die Gewichtungen müssen zusammen 100 % ergeben.",
			"details": [
			{
			"field": "positions",
			"message": "Summe der Gewichtungen ist 95 %."
			}
			]
			}
			}</pre>

GET	`/portfolios`
	Eigene Portfolios auflisten · Sprint 1
		Kein Body. Ab Sprint 2 Header: Authorization: Bearer <token>	**200 OK**<pre>{
			"portfolios": [
			{
			"id": 5,
			"name": "Mein Weltportfolio",
			"createdAt": "2026-09-30T10:12:00Z",
			"updatedAt": "2026-09-30T10:12:00Z",
			"positions": [
			{
			"assetId": 17,
			"symbol": "EUNL",
			"name": "iShares Core MSCI World",
			"weight": 70
			},
			{
			"assetId": 23,
			"symbol": "IS3N",
			"name": "iShares Core MSCI EM IMI",
			"weight": 30
			}
			]
			}
			]
			}</pre>

Method	Ressource	Data In	Data Out
GET	`/portfolios/{portfolioId}` Ein Portfolio abrufen · Sprint 1	Pfad: portfolioId, z. B. /portfolios/5. Kein Body.	**200 OK** `{ "id": 5, "name": "Mein Weltportfolio", "createdAt": "2026-09-30T10:12:00Z", "updatedAt": "2026-09-30T10:12:00Z", "positions": [{ "assetId": 17, "symbol": "EUNL", "name": "iShares Core MSCI World", "weight": 70 }, { "assetId": 23, "symbol": "IS3N", "name": "iShares Core MSCI EM IMI", "weight": 30 }] }` **404 Not Found** `{ "error": { "code": "NOT_FOUND", "message": "Portfolio nicht gefunden." } }`
PUT	`/portfolios/{portfolioId}` Portfolio ändern · Sprint 2	Pfad: portfolioId · Header: Authorization: Bearer <token> `{ "name": "Weltportfolio 80/20", "positions": [ { "assetId": 17, "weight": 80 }, { "assetId": 23, "weight": 20 } ] }`	**200 OK** `{ "id": 5, "name": "Weltportfolio 80/20", "createdAt": "2026-09-30T10:12:00Z", "updatedAt": "2026-10-06T08:00:00Z", "positions": [ { "assetId": 17, "symbol": "EUNL", "name": "iShares Core MSCI World", "weight": 80 }, { "assetId": 23, "symbol": "IS3N", "name": "iShares Core MSCI EM IMI",`

Method	Ressource	Data In	Data Out

			"weight": 20
			}
			]
			}
			400 Bad Request
			{
			"error": {
			"code": "VALIDATION_ERROR",
			"message": "Die Gewichtungen müssen zusammen 100 % ergeben."
			}
			}
			404 Not Found
			{
			"error": {
			"code": "NOT_FOUND",
			"message": "Portfolio nicht gefunden."
			}
			}

DELETE	`/portfolios/{portfolioId}`
	Portfolio löschen · Sprint 2
		Pfad: portfolioId · Header: Authorization: Bearer <token>. Kein Body.	**204 No Content**
			Kein Body
			404 Not Found
			{
			"error": {
			"code": "NOT_FOUND",
			"message": "Portfolio nicht gefunden."
			}
			}

Simulationen (US-04, US-05, US-08)

Method	Ressource	Data In	Data Out

POST	`/simulations`
	Simulation berechnen, optional speichern · Sprint 1–3
		{
		"portfolioId": 5,
		"parameters": {
		"startCapital": 10000,
		"monthlyContribution": 200,
		"horizonYears": 20,
		"paths": 10000,
		"seed": 42,
		"targetAmount": 100000
		},
			200 OK (save: false) bzw. 201 Created (save: true)
			{
			"id": null,
			"name": null,
			"portfolioId": 5,
			"parameters": {
			"startCapital": 10000,
			"monthlyContribution": 200,
			"horizonYears": 20,
			"paths": 10000,

Method	Ressource	Data In	Data Out

		<pre> "save": false,
		  "name": null
		}</pre>Pflicht: portfolioId, startCapital, monthlyContribution, horizonYears. Optional: paths (Standard 10.000), seed, targetAmount. save: true speichert (ab Sprint 3, dann ist name Pflicht).
			<pre> "seed": 42,
			    "targetAmount": 100000
			  },
			  "estimates": {
			    "historyFrom": "2016-09-01",
			    "historyTo": "2026-08-31",
			    "assets": [
			      {
			        "assetId": 17,
			        "annualReturn": 0.078,
			        "annualVolatility": 0.152
			      },
			      {
			        "assetId": 23,
			        "annualReturn": 0.051,
			        "annualVolatility": 0.183
			      }
			    ]
			  },
			  "timeline": [
			    {
			      "month": 0,
			      "p5": 10000,
			      "p50": 10000,
			      "p95": 10000
			    },
			    {
			      "month": 1,
			      "p5": 9740.1,
			      "p50": 10265.3,
			      "p95": 10820.7
			    },
			    {
			      "month": 240,
			      "p5": 61200.1,
			      "p50": 112340.55,
			      "p95": 205880.9
			    }
			  ],
			  "metrics": {
			    "totalContributions": 58000,
			    "medianEndValue": 112340.55,
			    "p5EndValue": 61200.1,
			    "p95EndValue": 205880.9,
			    "valueAtRisk95": 0,
			    "maxDrawdownMedian": 0.31,
			    "targetProbability": 0.587
			  },</pre>

Method	Ressource	Data In	Data Out

			<pre> "runtimeMs": 840,
			  "createdAt": "2026-09-30T10:15:00Z"
			}</pre>timeline enthält einen Punkt pro Monat (Monat 0 bis horizonYears × 12); hier gekürzt auf drei Punkte.
			metrics wird ab US-05 (Sprint 2) befüllt.
			422 Unprocessable Entity
			<pre>{
			  "error": {
			    "code": "INSUFFICIENT_DATA",
			    "message": "Für eine Anlage liegen weniger als 3 Jahre Kursdaten vor.",
			    "details": [
			      {
			        "field": "assetId",
			        "message": "23"
			      }
			    ]
			  }
			}</pre>**400 / 404 / 503**
			Ungültige Parameter / Portfolio nicht gefunden / Kursdaten nicht verfügbar

GET	`/simulations`
	Gespeicherte Simulationen · Sprint 3
		Header: Authorization: Bearer <token>. Kein Body.	**200 OK**
			<pre>{
			  "simulations": [
			    {
			      "id": 12,
			      "name": "Weltportfolio 20 Jahre",
			      "portfolioId": 5,
			      "createdAt": "2026-10-20T09:00:00Z"
			    }
			  ]
			}</pre>

GET	`/simulations/{simulationId}`
	Gespeicherte Simulation abrufen · Sprint 3
		Pfad: simulationId · Header: Authorization: Bearer <token>	**200 OK**
			Body wie bei POST /simulations, mit id und name, ohne Neuberechnung:
			<pre>{
			  "id": 12,
			  "name": "Weltportfolio 20 Jahre",
			  "portfolioId": 5,
			  "parameters": {
			    "startCapital": 10000,
			    "monthlyContribution": 200,</pre>

Method	Ressource	Data In	Data Out

			"horizonYears": 20,
			"paths": 10000,
			"seed": 42,
			"targetAmount": 100000
			},
			"estimates": {
			"historyFrom": "2016-09-01",
			"historyTo": "2026-08-31",
			"assets": [
			{
			"assetId": 17,
			"annualReturn": 0.078,
			"annualVolatility": 0.152
			},
			{
			"assetId": 23,
			"annualReturn": 0.051,
			"annualVolatility": 0.183
			}
			]
			},
			"timeline": [
			{
			"month": 0,
			"p5": 10000,
			"p50": 10000,
			"p95": 10000
			},
			{
			"month": 240,
			"p5": 61200.1,
			"p50": 112340.55,
			"p95": 205880.9
			}
			],
			"metrics": {
			"totalContributions": 58000,
			"medianEndValue": 112340.55,
			"p5EndValue": 61200.1,
			"p95EndValue": 205880.9,
			"valueAtRisk95": 0,
			"maxDrawdownMedian": 0.31,
			"targetProbability": 0.587
			},
			"runtimeMs": 840,
			"createdAt": "2026-09-30T10:15:00Z"
			}

Method	Ressource	Data In	Data Out
			timeline hier auf zwei Punkte gekürzt. **404 Not Found** `{ "error": { "code": "NOT_FOUND", "message": "Simulation nicht gefunden." } }`
DELETE	`/simulations/{simulationId}` Gespeicherte Simulation löschen · Sprint 3	Pfad: simulationId · Header: Authorization: Bearer <token>	**204 No Content** Kein Body **404 Not Found** `{ "error": { "code": "NOT_FOUND", "message": "Simulation nicht gefunden." } }`

Vergleiche (US-07, US-08, US-10)

Method	Ressource	Data In	Data Out
POST	`/comparisons` Vergleich berechnen, optional speichern · Sprint 3–4	`{ "items": [ { "label": "Mein Weltportfolio", "portfolioId": 5 }, { "label": "Nur S&P 500", "assetId": 31 } ], "parameters": { "startCapital": 10000, "monthlyContribution": 200, "horizonYears": 20, "paths": 10000, "seed": 42 }, "save": false, "name": null }`	**200 OK bzw. 201 Created** `{ "id": null, "name": null, "parameters": { "startCapital": 10000, "monthlyContribution": 200, "horizonYears": 20, "paths": 10000, "seed": 42 }, "results": [ { "label": "Mein Weltportfolio", "portfolioId": 5, "timeline": [ { "month": 0, "p5": 10000, "p50": 10000,`

Method	Ressource	Data In	Data Out

		Jedes Item hat entweder portfolioId oder assetId. 2 Items (US-07), ab US-10 bis zu 4. Alle Items werden mit denselben Zufallszahlen berechnet.	<pre>"p95": 10000
			},
			{
			"month": 240,
			"p5": 61200.1,
			"p50": 112340.55,
			"p95": 205880.9
			}
			],
			"metrics": {
			"totalContributions": 58000,
			"medianEndValue": 112340.55,
			"p5EndValue": 61200.1,
			"p95EndValue": 205880.9,
			"valueAtRisk95": 0,
			"maxDrawdownMedian": 0.31,
			"targetProbability": 0.587
			}
			},
			{
			"label": "Nur S&P 500",
			"assetId": 31,
			"timeline": [
			{
			"month": 0,
			"p5": 10000,
			"p50": 10000,
			"p95": 10000
			},
			{
			"month": 240,
			"p5": 55800,
			"p50": 121900.4,
			"p95": 248300.7
			}
			],
			"metrics": {
			"totalContributions": 58000,
			"medianEndValue": 121900.4,
			"p5EndValue": 55800,
			"p95EndValue": 248300.7,
			"valueAtRisk95": 0,
			"maxDrawdownMedian": 0.38,
			"targetProbability": 0.62
			}
			}
			],
			"pairwise": [</pre>

Method	Ressource	Data In	Data Out

			{
			"a": 0,
			"b": 1,
			"probabilityAOutperformsB": 0.46
			}
			],
			"runtimeMs": 1620,
			"createdAt": "2026-10-20T09:00:00Z"
			}
			probabilityAOutperformsB: Anteil der Pfade, in denen results[a] am Ende mehr wert ist als results[b].
			400 Bad Request
			{
			"error": {
			"code": "VALIDATION_ERROR",
			"message": "Bitte genau zwei Einträge zum Vergleich auswählen."
			}
			}

GET	`/comparisons`
	Gespeicherte Vergleiche auflisten · Sprint 3
		Header: Authorization: Bearer <token>. Kein Body.	**200 OK**
			{
			"comparisons": [
			{
			"id": 7,
			"name": "Welt vs. S&P 500",
			"itemLabels": [
			"Mein Weltportfolio",
			"Nur S&P 500"
			],
			"createdAt": "2026-10-20T09:00:00Z"
			}
			]
			}

GET	`/comparisons/{comparisonId}`
	Gespeicherten Vergleich abrufen · Sprint 3
		Pfad: comparisonId, z. B. /comparisons/7 · Header: Authorization: Bearer <token>. Kein Body.	**200 OK**
			Body wie bei POST /comparisons, mit id und name, ohne Neuberechnung:
			{
			"id": 7,
			"name": "Welt vs. S&P 500",
			"parameters": {
			"startCapital": 10000,
			"monthlyContribution": 200,
			"horizonYears": 20,
			"paths": 10000,

Method	Ressource	Data In	Data Out

			"seed": 42
			},
			"results": [
			{
			"label": "Mein Weltportfolio",
			"portfolioId": 5,
			"timeline": [
			{
			"month": 0,
			"p5": 10000,
			"p50": 10000,
			"p95": 10000
			},
			{
			"month": 240,
			"p5": 61200.1,
			"p50": 112340.55,
			"p95": 205880.9
			}
			],
			"metrics": {
			"medianEndValue": 112340.55
			}
			},
			{
			"label": "Nur S&P 500",
			"assetId": 31,
			"timeline": [
			{
			"month": 0,
			"p5": 10000,
			"p50": 10000,
			"p95": 10000
			},
			{
			"month": 240,
			"p5": 55800,
			"p50": 121900.4,
			"p95": 248300.7
			}
			],
			"metrics": {
			"medianEndValue": 121900.4
			}
			}
			],
			"pairwise": [
			{

Method	Ressource	Data In	Data Out

			"a": 0,
			"b": 1,
			"probabilityAOutperformsB": 0.46
			}
			],
			"runtimeMs": 1620,
			"createdAt": "2026-10-20T09:00:00Z"
			}
			timeline und metrics hier gekürzt; vollständig wie bei POST /comparisons.
			404 Not Found
			{
			"error": {
			"code": "NOT_FOUND",
			"message": "Vergleich nicht gefunden."
			}
			}

DELETE	`/comparisons/{comparisonId}`
	Gespeicherten Vergleich löschen · Sprint 3
		Pfad: comparisonId · Header: Authorization: Bearer <token>. Kein Body.	**204 No Content**
			Kein Body
			404 Not Found
			{
			"error": {
			"code": "NOT_FOUND",
			"message": "Vergleich nicht gefunden."
			}
			}

Gastmodus (US-09)

Method	Ressource	Data In	Data Out
GET	`/demo/portfolios` Beispielportfolios · Sprint 4	Kein Body, keine Anmeldung	**200 OK** { "portfolios": [{ "id": 900, "name": "Beispiel: Weltportfolio", "positions": [{ "assetId": 17, "symbol": "EUNL", "name": "iShares Core MSCI World", "weight": 100

Method	Ressource	Data In	Data Out
			<pre> }] }] }</pre>

POST	`/demo/simulations`
	Beispielsimulation ohne Speicherung · Sprint 4
		<pre>{
		  "portfolioId": 900,
		  "parameters": {
		    "startCapital": 5000,
		    "monthlyContribution": 100,
		    "horizonYears": 15
		  }
		}</pre>paths höchstens 10.000; kein save und name.
			200 OK
			Body wie bei POST /simulations, id ist immer null:
			<pre>{
			  "id": null,
			  "portfolioId": 900,
			  "parameters": {
			    "startCapital": 5000,
			    "monthlyContribution": 100,
			    "horizonYears": 15,
			    "paths": 10000,
			    "seed": 731
			  },
			  "timeline": [
			    {
			      "month": 0,
			      "p5": 5000,
			      "p50": 5000,
			      "p95": 5000
			    },
			    {
			      "month": 180,
			      "p5": 24100.3,
			      "p50": 41250.8,
			      "p95": 70980.2
			    }
			  ],
			  "metrics": {
			    "totalContributions": 23000,
			    "medianEndValue": 41250.8,
			    "p5EndValue": 24100.3,
			    "p95EndValue": 70980.2,
			    "valueAtRisk95": 0,
			    "maxDrawdownMedian": 0.29,
			    "targetProbability": null
			  },
			  "runtimeMs": 610,
			  "createdAt": "2026-11-10T12:00:00Z"
			}</pre>